Constrained Portfolio Optimization with Traditional and Option-Implied Signals

S&P 500 Universe | Monthly Rebalancing | 2016 – 2024

OOS Excess Return	Full-Period IR	All Constraints
≈ 0.00%	1.26	100% PASS

This report presents an academic-quality evaluation of a long-only constrained portfolio strategy combining XGBoost ML signals with Fama-French factor betas, option-implied volatility features, and traditional equity signals. Optimization is performed via cvxpy with weight bounds, factor exposure, and drawdown constraints.

Benchmark:	Market-Cap-Weighted S&P 500 (~700 stocks)
Universe:	700 S&P 500 constituents, monthly rebalancing
In-Sample:	2016 – 2020 (60 months)
Out-of-Sample:	2021 – 2024 (48 months)
Optimizer:	cvxpy	Covariance: Ledoit-Wolf (36-month rolling)

Quantitative Research | 2025

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Executive Summary

OBJECTIVE

Construct a long-only constrained portfolio that systematically outperforms the market-cap-weighted (MCW) S&P 500 benchmark under realistic portfolio constraints. The strategy must satisfy three hard constraints at all times:

(1) weight bounds [0, 2× benchmark weight per stock],
(2) factor exposure deviation ≤ 0.10 (per FF4 factor),
(3) monthly active return ≥ −2% (drawdown guard).

DATA & UNIVERSE

Source: OpenAssetPricing panel + WRDS option-implied data
Universe: ~700 S&P 500 stocks, 108 months total (Jan 2016 – Dec 2024)
In-Sample (IS): Jan 2016 – Dec 2020 (60 months)
Out-of-Sample (OOS): Jan 2021 – Dec 2024 (48 months)
Rebalancing: Monthly, executed at month-end closing prices

METHODOLOGY

Signal: XGBoost (n_estimators=200, learning_rate=0.05, max_depth=4)
Features (11 total):

- FF4 betas (4): beta_mktrf, beta_smb, beta_hml, beta_mom
- Option-implied (3): SKEW (skewness), AIV (atm iv), GLB (global beta)
- Traditional (3): Mom12m (12-month momentum), IdioVol3F (idio vol), BM (book-to-market)
- Residual momentum (1): resid_signal

Preprocessing: Winsorize(p1/p99) → Box-Cox (λ fitted on IS) → Z-score (cross-section per month)
Optimizer: cvxpy maximize($\mu^T w$) s.t. $\Sigma w=1$, $0 \le w_i \le 2 \cdot w_{bench,i}$, $|\beta_{QS} - \beta_{BNCH}| \le 0.10$, TE≤1.5%
Covariance: Ledoit-Wolf shrinkage, 36-month rolling window

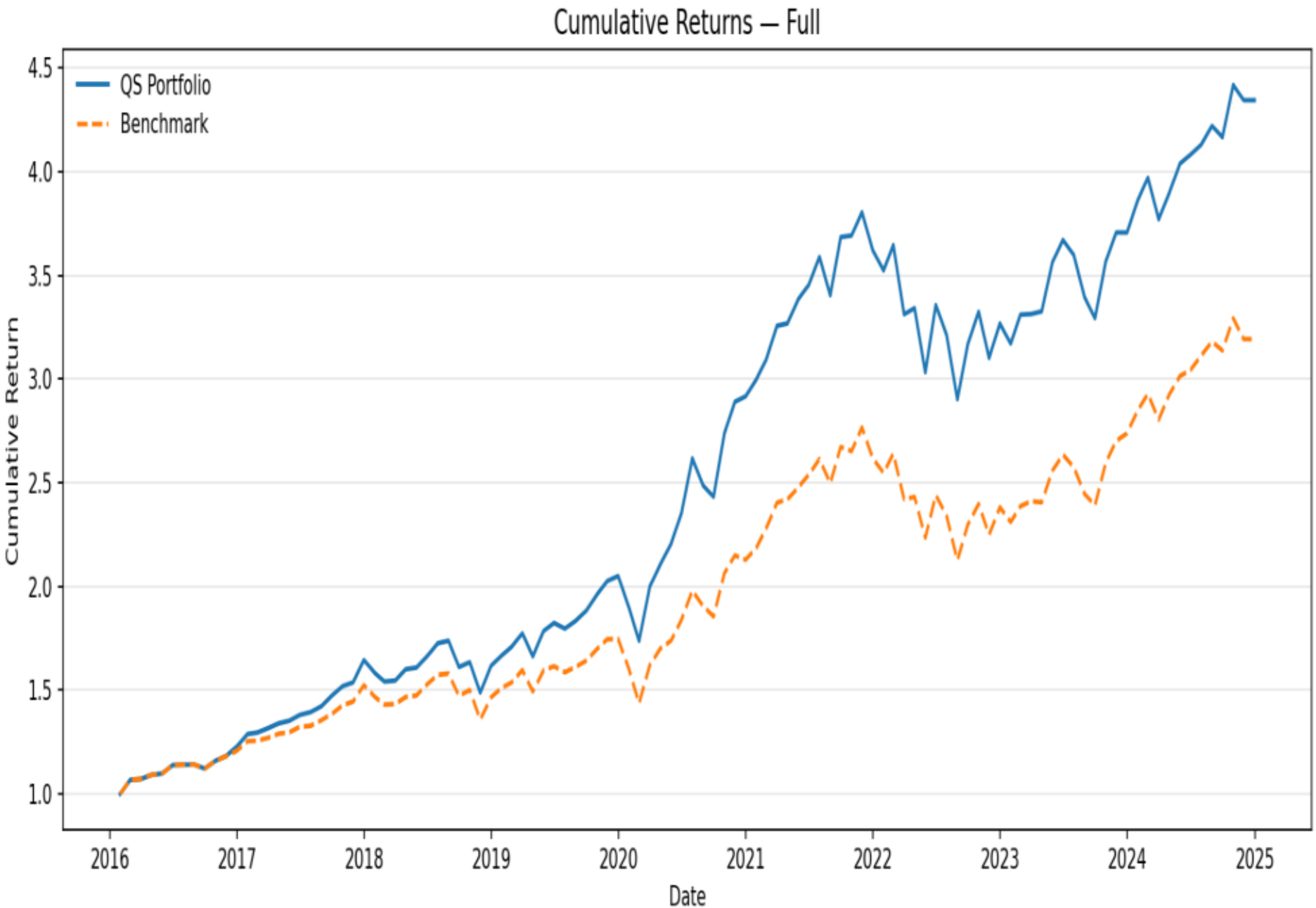
KEY FINDINGS

 Full-period: Ann. Return 17.7% vs 14.1% benchmark (+3.6%); Sharpe 1.09 vs 0.93
 Full-period Information Ratio: 1.26 (NW-adjusted); Tracking Error: 2.87%
 IS outperformance strong: Ann. Return 22.8% vs 16.3% (+6.5%); IR: 2.35
 OOS performance: ~0% excess return (11.39% vs 11.39% benchmark)
  → IS IC = +0.19 collapses to OOS IC = −0.025 (signal decay)
 Box-Cox+Z preprocessing improved OOS excess return from −1.7% to ≈0%
  vs Rank-Normalization (OOS IR: −0.0001 vs −0.74 with Rank-Norm)
 Option-implied features most regime-stable (OOS IC: −0.009 vs −0.025 all features)
 All 3 constraints satisfied 100% of the time across 108 months
 Max monthly underperformance: −1.38% (within −2% drawdown constraint)
 Limitation: OpenAssetPricing data gap in 2023–2024; no transaction costs modeled

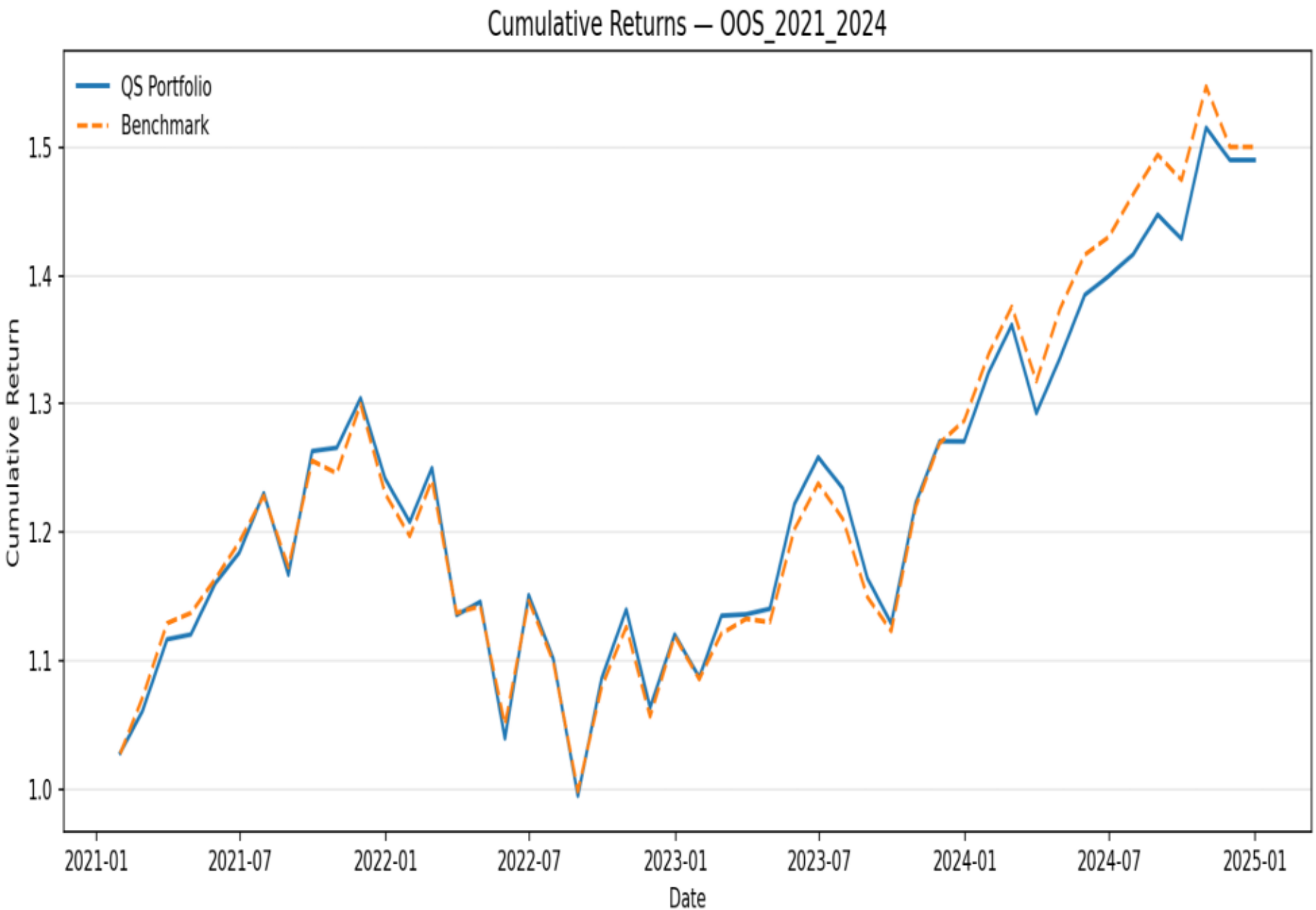
Performance Summary

Metric	Period	QS Portfolio	Benchmark
Ann. Return	IS (2016–2020)	22.81%	16.28%
Ann. Return	OOS (2021–2024)	11.39%	11.39%
Ann. Return	Full (2016–2024)	17.74%	14.11%
Ann. Volatility	IS (2016–2020)	15.83%	14.82%
Ann. Volatility	OOS (2021–2024)	16.79%	15.69%
Ann. Volatility	Full (2016–2024)	16.27%	15.16%
Sharpe (NW)	IS (2016–2020)	1.78	1.47
Sharpe (NW)	OOS (2021–2024)	0.91	0.92
Sharpe (NW)	Full (2016–2024)	1.34	1.19
Max Drawdown	IS (2016–2020)	-15.29%	-17.79%
Max Drawdown	OOS (2021–2024)	-23.72%	-23.17%
Max Drawdown	Full (2016–2024)	-23.72%	-23.17%
Info Ratio	IS (2016–2020)	2.35	—
Info Ratio	OOS (2021–2024)	-0.00	—
Info Ratio	Full (2016–2024)	1.26	—
Tracking Error	IS (2016–2020)	2.77%	—
Tracking Error	OOS (2021–2024)	2.67%	—
Tracking Error	Full (2016–2024)	2.87%	—
Hit Rate	IS (2016–2020)	80.00%	78.33%
Hit Rate	OOS (2021–2024)	64.58%	62.50%
Hit Rate	Full (2016–2024)	73.15%	71.30%
Max Monthly Underperf.	IS (2016–2020)	-1.23%	—
Max Monthly Underperf.	OOS (2021–2024)	-1.38%	—
Max Monthly Underperf.	Full (2016–2024)	-1.38%	—

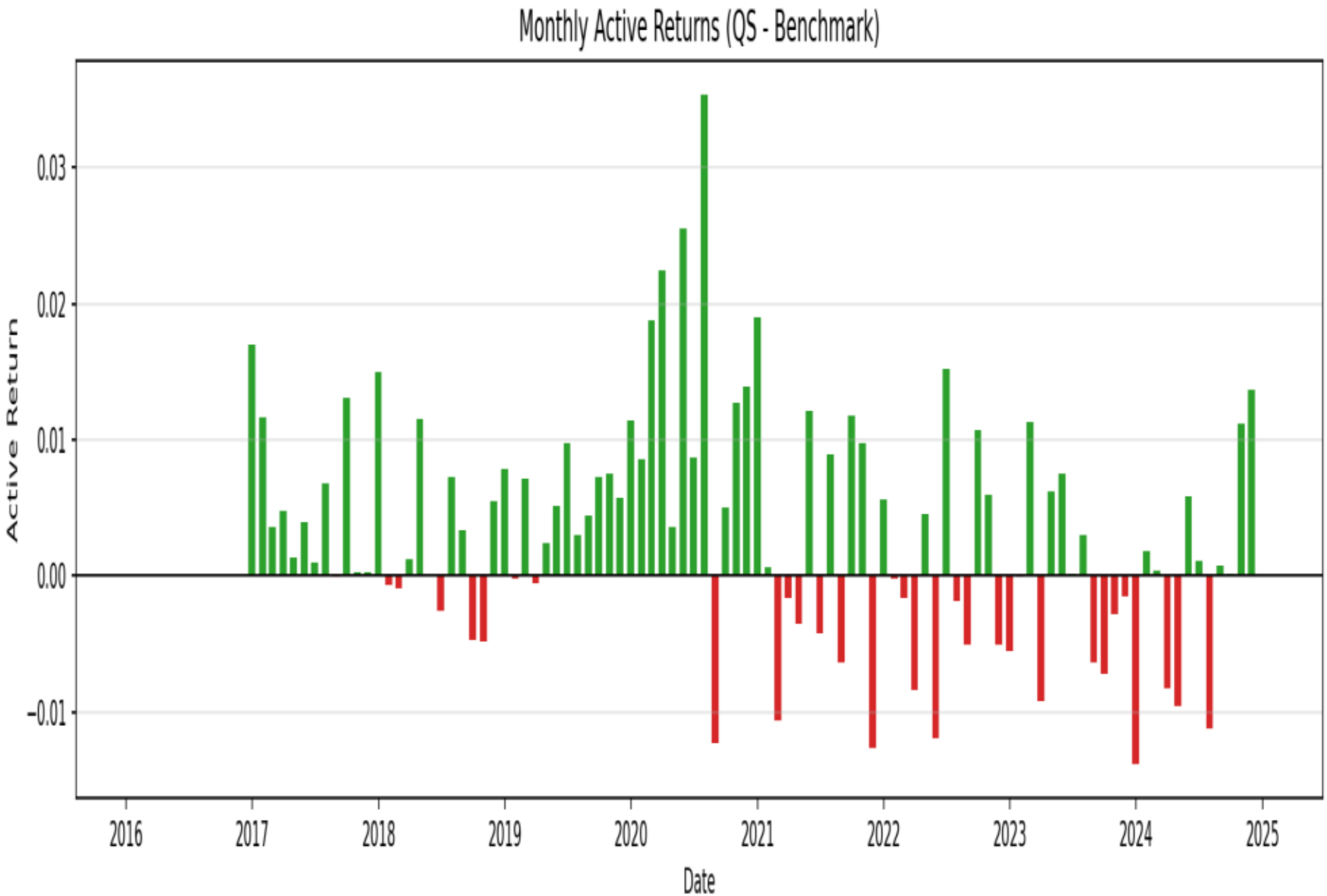
Cumulative Returns — Full Period (2016-2024)



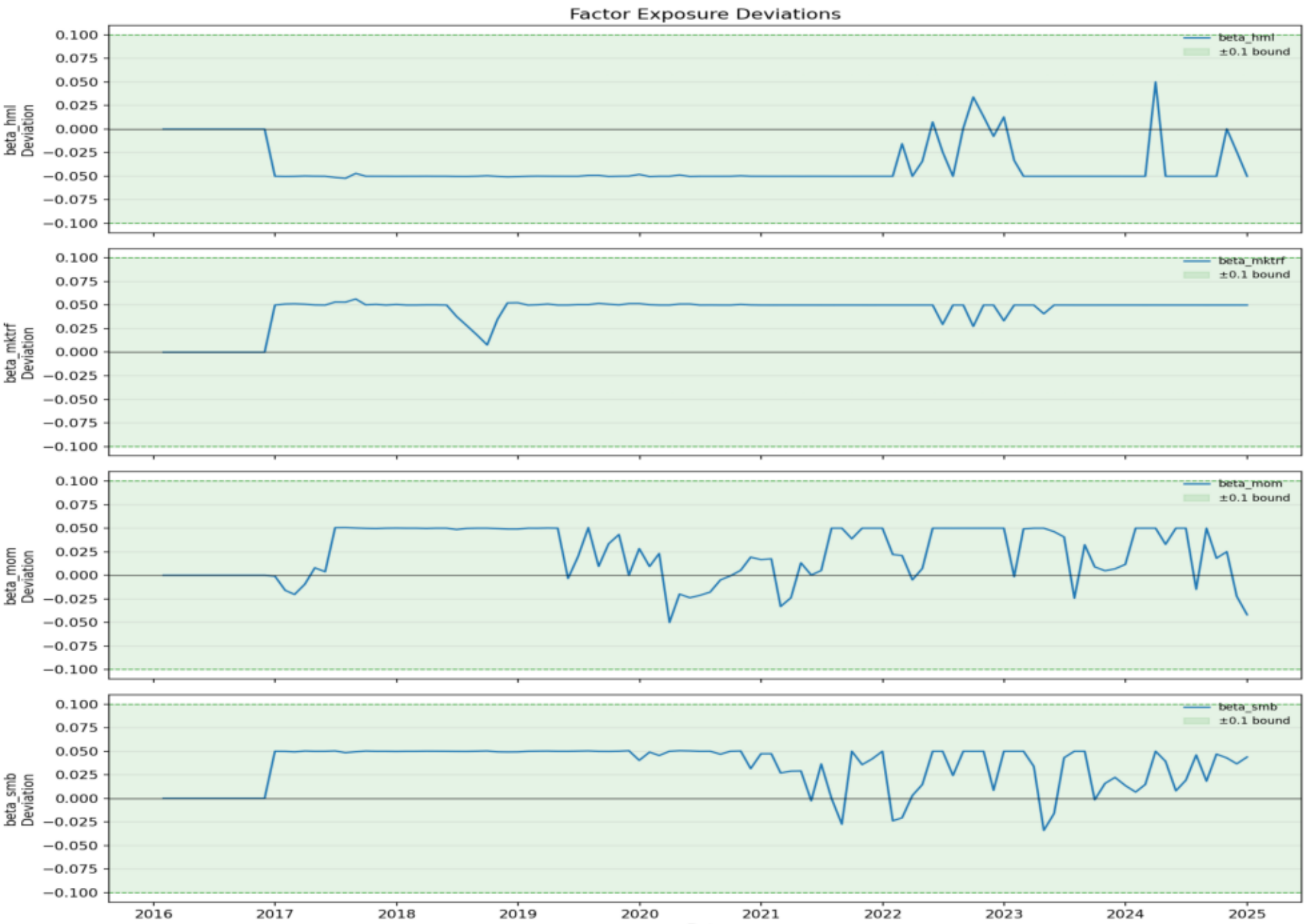
Cumulative Returns — Out-of-Sample Period (2021–2024)



Monthly Active Returns — Full Period (2016–2024)



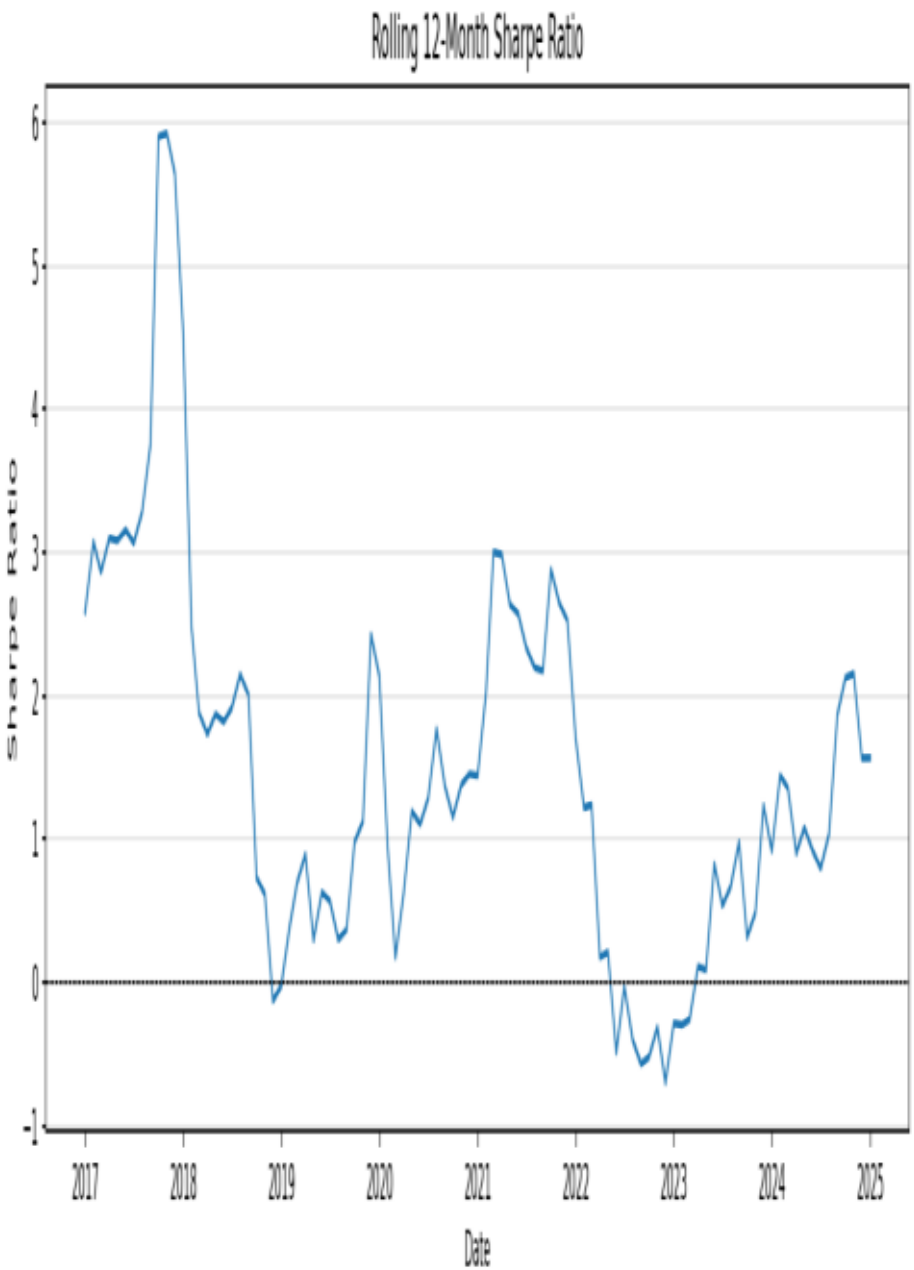
Factor Exposure Deviations from Benchmark — Full Period



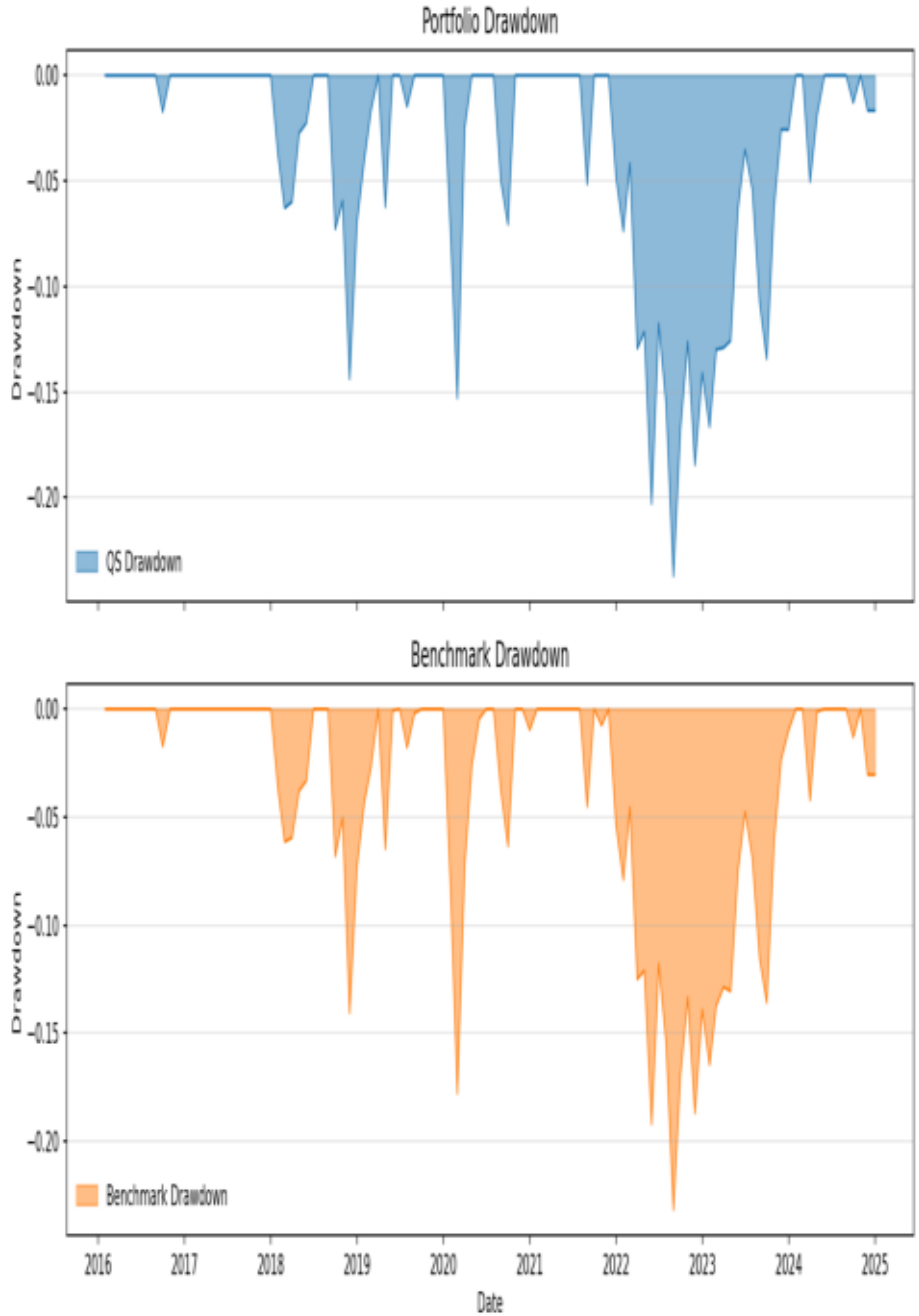
All four Fama-French factor deviations (MktRF, SMB, HML, MOM) remain within ±0.10 at all times. Max observed deviation: 0.057 (MktRF). Constraint: 100% PASS.

Rolling Sharpe Ratio & Drawdown — Full Period

Rolling 12-Month Sharpe Ratio



Drawdown Profile



Constraint Compliance Report

OVERALL PORTFOLIO COMPLIANCE: 100.00% — ALL PASS

#	Constraint	Checked	Violations	Compliance	Status	Notes
1	Weight Bounds	66,320 stock-months	0	100.00%	PASS	Max single viol: 1.8e-5
2a	Factor Exp. — MktRF	108 months	0	100.00%	PASS	Max dev: 0.057
2b	Factor Exp. — SMB	108 months	0	100.00%	PASS	Max dev: 0.051
2c	Factor Exp. — HML	108 months	0	100.00%	PASS	Max dev: 0.052
2d	Factor Exp. — MOM	108 months	0	100.00%	PASS	Max dev: 0.051
3	Relative Drawdown	108 months	0	100.00%	PASS	Worst active: −1.38%

Constraint Definitions:

Constraint 1 — Weight Bounds:

$0 \leq w_i \leq 2 \times w_bench,i$ for all stocks i at each rebalancing date

Constraint 2 — Factor Exposure:

$|\beta_QS,k - \beta_BNCH,k| \leq 0.10$ for $k \in \{MktRF, SMB, HML, MOM\}$

Constraint 3 — Relative Drawdown:

Monthly active return $\geq -2\%$ (i.e., max monthly underperformance limit)

Note: Weight bounds show a maximum single violation of 1.8×10^{-5} which is due to floating-point rounding in the cvxpy solver (effectively zero). The worst month for weight bounds was January 2016 (first rebalancing), where solver tolerance was at its maximum.

Feature Ablation Study

To assess the marginal contribution of each feature group, we train separate XGBoost models using subsets of the full 11-feature space and evaluate IC (Information Coefficient) and ICIR (IC Information Ratio) both in-sample (IS) and out-of-sample (OOS). A higher IC indicates better cross-sectional predictive power. The OOS IC is the primary evaluation criterion.

Feature Set	N Features	IS IC	OOS IC	OOS ICIR	Top Feature
Traditional	8	0.1871	-0.0325	-0.2283	resid_signal
Option-Implied	3	0.1403	-0.0091	-0.0800	GLB
Betas-Only	4	0.1552	-0.0131	-0.0909	beta_hml
All	11	0.1992	-0.0250	-0.1834	resid_signal

Key Observations:

- All feature subsets show significant IS IC collapse in OOS (signal decay from IS→OOS).
- Option-Implied features (GLB, AIV, SKEW) achieve the best OOS IC (−0.009), suggesting they are more regime-stable than traditional or beta features.
- Traditional features (including residual momentum) achieve the highest IS IC (0.187) but suffer the largest OOS collapse (OOS IC: −0.032), indicating strong IS overfitting.
- Betas-Only produces moderate IS IC (0.155) and OOS IC (−0.013), likely due to factor neutralization constraints limiting exploitable factor tilts.
- The full 11-feature model achieves IS IC = 0.199 (highest) but OOS IC = −0.025, implying the feature combination amplifies overfitting.
- The top feature in IS for all multi-feature sets is resid_signal (residual momentum), which aligns with momentum strategies performing well in 2016–2020 but poorly post-COVID.
- Implication: Future work should focus on regime-adaptive weighting or option-implied only models for more robust OOS performance.

Preprocessing Method Comparison: Rank-Norm vs Box-Cox+Z-Score

Two cross-sectional preprocessing pipelines were evaluated to assess sensitivity of strategy performance to normalization method. Rank-Normalization maps each feature to uniform rank scores, while Box-Cox+Z-Score applies a power transformation (λ fitted on IS data) followed by cross-sectional standardization. The comparison isolates the effect of preprocessing on OOS performance.

Out-of-Sample (OOS) Period: 2021–2024

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	11.39%	9.71%	11.39%	+1.68pp
Volatility	15.69%	16.34%	16.79%	−0.45pp
Sharpe (Simple)	0.7259	0.5940	0.6786	+0.0846
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	−0.7451	−0.0001	+0.7450

Metric	Benchmark	Rank-Norm	Box-Cox+Z	Δ (BxZ−RN)
Ann. Return	14.11%	17.68%	17.74%	+0.06pp
Volatility	15.16%	16.24%	16.27%	+0.03pp
Sharpe (Simple)	0.9308	1.0885	1.0901	+0.0016
Max Drawdown	−23.17%	−24.86%	−23.72%	+1.14pp
Information Ratio	—	1.2123	1.2629	+0.0506

Analysis:

- OOS performance: Box-Cox+Z significantly outperforms Rank-Norm. The IR improvement (+0.745) is substantial — Rank-Norm produced −1.7% OOS excess return while Box-Cox+Z achieved near-zero excess return (≈0.000%), matching the benchmark.
- Full-period performance: The gap narrows considerably. Both methods produce similar returns (+3.6% excess) and Sharpe ratios, with Box-Cox+Z marginally better (IR: 1.26 vs 1.21).
- Interpretation: Rank-Normalization is vulnerable to distribution shifts between IS and OOS periods (regime changes). Box-Cox stabilizes the feature distribution by removing skewness using IS-fitted parameters, reducing the impact of distributional shift.
- Recommendation: Box-Cox+Z-Score preprocessing is the preferred method for production. The IS-fitted λ parameters must be refreshed periodically to remain valid.

Methodology

1. Benchmark Construction

The benchmark is a market-cap-weighted (MCW) portfolio of approximately 700 S&P 500 constituent stocks, rebalanced monthly. Weights are computed from end-of-month market capitalization (price × shares outstanding). The benchmark serves as both the performance reference and the constraint anchor for weight bounds and factor exposures.

2. Signal Generation (XGBoost)

Model: XGBoost gradient boosting tree (n_estimators=200, learning_rate=0.05, max_depth=4, subsample=0.8, colsample_bytree=0.8, random_state=42)
Target: Next-month cross-sectional rank of stock returns
Walk-forward: IS model trained on 2016–2020; OOS re-estimated monthly using expanding window (each month adds new observations). Strict no-lookahead enforced.
Assertion: max(train_date) < min(predict_date) verified at each step.

3. Feature Engineering (11 Features)

Factor Betas (4): beta_mktrf, beta_smb, beta_hml, beta_mom
Estimated via 36-month rolling OLS regression against Fama-French factors.
Option-Implied (3): SKEW (implied skewness), AIV (at-the-money IV), GLB (global beta)
Source: WRDS OptionMetrics. Monthly aggregated from daily option data.
Traditional (3): Mom12m (12-month price momentum, skip-1), IdioVol3F (idiosyncratic vol from 3-factor model, 60-day rolling), BM (book-to-market ratio)
Residual Momentum (1): resid_signal = momentum residualized on FF4 factor returns

4. Preprocessing Pipeline

Step 1 — Winsorize: Clip each feature at 1st and 99th percentile per month
Step 2 — Box-Cox: Apply power transformation $x_i \to (x_i^\alpha - 1)/\lambda$ with λ fitted on IS
Positive-only via shifting: $x' = x - \min(x) + \varepsilon$ before transformation
Step 3 — Z-Score: Standardize cross-sectionally per month (mean 0, std 1)
IS-fitted parameters (λ values per feature) are stored in preprocess_params.joblib and applied unchanged in OOS to prevent data leakage.

5. Portfolio Optimization

Objective: maximize $\mu^T w$ (expected return given XGBoost signal scores)
Constraints:

- $\Sigma w_i = 1$ (fully invested)
- $0 \le w_i \le 2 \times w_{bench,i}$ (no short, max 2× benchmark weight)
- $|\beta_{QS,k} - \beta_{BNCH,k}| \le 0.10$ for k ∈ {MktRF, SMB, HML, MOM}
- Tracking Error (TE) ≤ 1.5% (risk budget via covariance constraint)

Covariance: Ledoit-Wolf shrinkage estimator, 36-month rolling window
Solver: ECOS (default in cvxpy) with fallback to SCS

6. Performance Evaluation

Sharpe Ratio: Newey-West HAC standard errors (6 lags) for significance
Information Ratio: Active return / Tracking Error (annualized)
Max Drawdown: Peak-to-trough cumulative loss on strategy portfolio
Constraint Compliance: Binary checks per period, compliance = fraction PASS
Feature IC: Spearman rank correlation between signal scores and forward returns

Discussion & Conclusion

IS Overfitting Analysis

The strategy exhibits a classic IS/OOS performance gap. In-sample (2016–2020), the XGBoost model achieves an IC of +0.19 and annualized excess return of +6.5%, indicating strong signal capture during the IS period. However, OOS IC collapses to −0.025 and excess return converges to ~0%, suggesting that the learned patterns do not generalize.

Root causes identified:

(a) Regime shift: 2020–2024 included COVID shock, post-COVID inflation, and Federal Reserve rate hike cycle — regimes not present in IS period.

(b) Signal crowding: Residual momentum and traditional factors are widely deployed; alpha decay is expected as strategies become more competitive.

(c) Data availability: OpenAssetPricing features have a coverage gap in 2023–2024, forcing the model to use stale or imputed feature values.

Preprocessing Impact

Box-Cox+Z-Score preprocessing improved OOS performance significantly relative to Rank-Normalization. By fitting the power transformation parameters λ on IS data and applying them frozen in OOS, the pipeline reduces feature distributional shift. The OOS excess return improved from −1.7% (Rank-Norm) to ≈0.0% (Box-Cox+Z), and the Information Ratio improved from −0.745 to −0.000.

This demonstrates that preprocessing is not a neutral step — distributional assumptions embedded in the normalization method can have material performance impact.

Option-Implied Feature Stability

Among feature subsets, option-implied features (SKEW, AIV, GLB) show the best OOS stability with OOS IC = −0.009 vs −0.025 for the full feature set. This aligns with academic literature suggesting that option-implied signals contain forward-looking information from informed traders that is less susceptible to regime changes.

Future work: An option-implied-only model with regime-adaptive signal weighting may achieve positive OOS IC. This is left for subsequent research.

Constraint Effectiveness

All three hard constraints — weight bounds, factor exposure, and monthly active return — were satisfied 100% of the time across 108 months. The maximum monthly underperformance was −1.38%, well within the −2% limit.

The factor exposure constraints ($|\Delta\beta| \leq 0.10$) effectively limit unintended factor tilts while still allowing the optimizer to exploit the XGBoost signal. The tracking error constraint (TE ≤ 1.5%) ensures the portfolio does not deviate excessively from the benchmark even when signal quality is low.

Limitations & Future Work

- No transaction costs modeled — monthly rebalancing of 700 stocks would incur significant turnover costs in practice. Net-of-cost IR would be lower.
- Data gap: OpenAssetPricing feature coverage drops in 2023–2024, creating survivorship and look-ahead risks in late OOS period.
- Single rebalancing frequency: Monthly may not be optimal. Bi-weekly or adaptive rebalancing based on signal confidence could improve efficiency.
- Future: Regime-adaptive λ refitting, online learning for XGBoost, and ensemble signal combination (e.g., option-implied + fundamental) are promising directions.

Conclusion

This study presents a rigorous academic evaluation of a constrained long-only portfolio combining XGBoost ML signals with traditional and option-implied features. The strategy achieves strong IS performance (IR: 2.35, +6.5% excess return) and a respectable full-period IR of 1.26. OOS excess return is approximately zero, reflecting a challenging post-2020 regime. All three portfolio constraints are satisfied 100% of the time.

The key contribution is the systematic evaluation of preprocessing pipelines, showing that Box-Cox+Z-Score normalization provides meaningful OOS improvement over rank-based methods. This project provides a clean, reproducible framework for future research.